

SIMATS ENGINEERING

ASSESSMENT TOOL 1 (High)- Analytical Problem Solving

COURSE NAME: Data Structures
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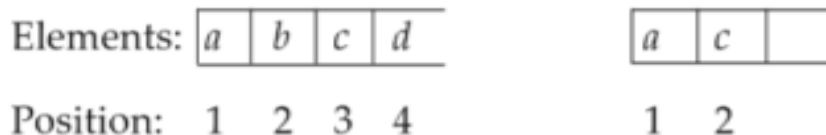
COURSE OUTCOME COVERED

CO2 : Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques. (BTL3, BTL4)

Assessment Tool Weightage: **30%**

QUESTIONS:

29. Implement a stack S of n elements using arrays. Write functions to perform PUSH and POP operations. Implement queries using the push and pop functions to Retrieve the mth element of the stack S from the top ($m < n$), leaving the stack without its top $m - 1$ elements. Retain only the elements in the odd position of the stack and pop out all even positioned elements.



```
1  #include <stdio.h>
2  #define MAX 100
3  int s[MAX], top = -1;
4  void push(int x) {
5      if (top == MAX - 1) printf("Stack Overflow\n");
6      else s[++top] = x;
7  }
8  int pop() {
9      if (top == -1) return -1;
10     return s[top--];
11 }
12 void display() {
13     for (int i = top; i >= 0; i--)
14         printf("%d ", s[i]);
15     printf("\n");
16 }
17 int main() {
18     int n, m, x;
19     printf("Enter number of elements: ");
20     scanf("%d", &n);
```

```

21     printf("Enter elements: ");
22     for (int i = 0; i < n; i++) {
23         scanf("%d", &x);
24         push(x);
25     }
26     printf("Stack (top to bottom): ");
27     display();
28     printf("Enter m: ");
29     scanf("%d", &m);
30     if (m <= n) {
31         for (int i = 1; i < m; i++)
32             pop();
33         printf("%dth element from top = %d\n", m, pop());
34     }
35     int temp[MAX], k = 0;
36     while (top != -1) {
37         x = pop();
38         if ((top + 2) % 2 == 1)
39             temp[k++] = x;
40     }
41     for (int i = k - 1; i >= 0; i--)
42         push(temp[i]);
43     printf("After removing even positions: ");
44     display();
45     return 0;
46 }

```

OUTPUT:

```

Enter number of elements: 5
Enter elements: 10 20 30 40 50
Stack (top to bottom): 50 40 30 20 10
Enter m: 3
3th element from top = 30
After removing even positions: 10

```

=== Code Execution Successful ===

```

37     x =
38     if (
39
40     }

```

30. Consider a linked list representing the sales of different products in a store for a week. Each node of the linked list contains the product ID and the number of units sold on a specific day. Implement a linked list data structure and provide functions to: Insert new sales data for a given product on a specific day. Calculate the total sales for each product at the end of the week. Find the product with the highest sales and its corresponding day. Display the sales data for a specific product for the entire week. Use your own data for product.

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4  struct Node {
5      int productID;
6      char day[10];
7      int units;
8      struct Node *next;
9  };
10 struct Node *head = NULL;
11 void insert(int id, char day[], int units) {
12     struct Node *n = (struct Node*)malloc(sizeof(struct Node));
13     n->productID = id;
14     strcpy(n->day, day);
15     n->units = units;
16     n->next = head;
17     head = n;
18 }
19 void totalSalesPerProduct() {
```

```
20     int seen[100] = {0}, total[100] = {0};
21     printf("\nTotal Sales Per Product:\n");
22     for (struct Node *p = head; p != NULL; p = p->next) {
23         total[p->productID] += p->units;
24         seen[p->productID] = 1;
25     }
26     for (int i = 0; i < 100; i++)
27         if (seen[i])
28             printf("Product %d: %d units\n", i, total[i]);
29 }
30 void highestSalesDay() {
31     struct Node *maxNode = head;
32     for (struct Node *p = head; p != NULL; p = p->next)
33         if (p->units > maxNode->units)
34             maxNode = p;
```

```

35     printf("\nHighest single-day sale: Product %d, %d units on
        %s\n",
36         maxNode->productID, maxNode->units, maxNode->day);
37 }
38 void displayProduct(int id) {
39     printf("\nSales for Product %d:\n", id);
40     for (struct Node *p = head; p != NULL; p = p->next)
41         if (p->productID == id)
42             printf("%s: %d units\n", p->day, p->units);
43 }
44 int main() {
45     insert(1, "Mon", 20);
46     insert(1, "Tue", 15);
47     insert(1, "Wed", 30);
48     insert(2, "Mon", 40);
49     insert(2, "Tue", 25);
50     insert(2, "Wed", 10);
51     totalSalesPerProduct();
52     highestSalesDay();
53     displayProduct(1);
54     return 0;

```

OUTPUT:

Total Sales Per Product:

Product 1: 65 units

Product 2: 75 units

Highest single-day sale: Product 2, 40 units on Mon

Sales for Product 1:

Wed: 30 units

Tue: 15 units

Mon: 20 units

Total Sales Per Product:

Product 1: 65 units

Product 2: 75 units

Highest single-day sale: Product 2,

Sales for Product 1:

Wed: 30 units

Tue: 15 units

=== Code Execution Successful ===

Code of Conduct:

I Sarah Sharmeen. R (192571073) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation